

TWO-POINT EXTREMALS FOR CHARACTERISTIC-FUNCTION DEPENDENCE WITH FIXED MARGINALS

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ABSTRACT. For random variables $X_1, \dots, X_d$ we study the largest characteristic-function defect $|\mathbb{E}e^{i\sum_j t_j X_j} - \prod_j \mathbb{E}e^{it_j X_j}|$. In every regime where we determine the exact constant, the extremal couplings make the phase sum two-point, and a certificate made of two tangent lines proves optimality. Without restrictions on the marginals, a companion paper establishes the sharp benchmark $C_d = \max_{0 \leq x \leq \pi} \{\cos^d(x/d) - \cos x\}$ ($C_3 = 2/\sqrt{3}$, $C_4 = 9/7$) from an exact profile in the moduli of the means. Here the marginal laws are prescribed. For a common symmetric unimodal law F we prove $K_d^F = \max_x \{\varphi(x/(d\mathbb{E}|X|))^d - \cos x\}$ under explicit conditions — for the uniform and triangular laws for all $d \geq 3$, for the B-spline and Epanechnikov laws for $d \geq 4$ — and show that, under strict versions of these conditions, attainment is equivalent to complete mixability of $|X|$; Gaussian marginals do not attain the bound. The uniform law is the least restrictive symmetric unimodal marginal for $d \geq 4$ but not for $d = 3$; without symmetry the law $\frac{1}{4}\delta_0 + \frac{3}{4}U[0, 1]$ beats it for every $d \geq 4$ and is asymptotically extremal among all unimodal laws, through the sharp inequality $\text{Var}X \geq \frac{9}{7}(\mathbb{E}|X - \text{med}X|)^2$. For every non-degenerate law with bounded support that is lattice or satisfies Cramér’s condition, $K_d^F = 2 - \pi^2 \text{Var}X / (2d(\mathbb{E}|X - \text{med}X|)^2) + O(d^{-3/2})$: the median replaces the centre, and no symmetry, unimodality or mixability is needed. Without symmetry the best two-point split is asymmetric; that it is optimal is conjectured. The finite verifications are short tables of explicit inequalities, checked by monotone brackets, and by interval arithmetic for the non-symmetric comparison.

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1. INTRODUCTION

For random variables $X_1, \dots, X_d$ and frequencies $t \in \mathbb{R}^d$ consider the *characteristic-function defect*

$$\Delta = \mathbb{E}e^{i\sum_j t_j X_j} - \prod_j \mathbb{E}e^{it_j X_j}.$$

It vanishes under independence, and its vanishing at all t on the diagonal is sub-independence; it is the basic quantity behind characteristic-function measures of dependence. We ask how large $|\Delta|$ can be, in three settings of decreasing freedom: (a) free laws with $|Z_j| \leq 1$ in place of $e^{it_j X_j}$; (b) prescribed moduli $|\mathbb{E}Z_j| = r_j$; (c) prescribed marginal laws $X_j \sim F$ (or F_j), with the coupling and the frequencies free.

Results. (a)–(b) are settled in the companion paper [1]: the sharp profile $C_d(\vec{r}) = \prod r_j - \cos(\min\{\sum \arccos r_j, \pi\})$ and the constant $C_d = \max_{0 \leq x \leq \pi} \{\cos^d(x/d) - \cos x\}$ (recalled as Theorem 3.1 and Corollary 3.2). (c): for symmetric unimodal F ,

$$K_d^F = \kappa_d^F := \max_x \{\varphi(x/(d\mathbb{E}|X|))^d - \cos x\}$$

under explicit conditions (Theorem 5.1), for all $d \geq 3$ for the uniform and triangular laws and for $d \geq 4$ for the B-spline and Epanechnikov laws (Table 1); the analogue for different laws holds under joint mixability (Theorem 5.6). Under strict versions of the hypotheses, attainment is equivalent to complete mixability of $|X|$ (Theorem 9.1); Gaussian marginals do not attain κ_d^G (Corollary 9.2).

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The uniform law has the largest constant among symmetric unimodal marginals for $d \geq 4$ but not for $d = 3$ (Theorems 8.3 and 8.7); without symmetry it is beaten for every $d \geq 4$ by $\frac{1}{4}\delta_0 + \frac{3}{4}U[0, 1]$, which is asymptotically extremal among all unimodal laws (Theorem 11.5, Corollary 11.3), by the sharp inequality $\text{Var}X \geq \frac{9}{7}(\mathbb{E}|X - \text{med}X|)^2$ for unimodal laws, a moment form of the known worst-case expected shortfall (Proposition 11.1). For every non-degenerate law with bounded support that is lattice or satisfies Cramér's condition, $K_d^F = 2 - \pi^2 \text{Var}X / (2d(\mathbb{E}|X - \text{med}X|)^2) + O(d^{-3/2})$ (Theorem 7.1); the factor $\text{Var}X / (\mathbb{E}|X - \text{med}X|)^2 \geq 1$ is the price of fixing the marginals. Without symmetry, the best two-point split is asymmetric; its optimality is Conjecture 10.2.

The two-point principle. In every exact regime proved here the extremal phase sum takes two values, and optimality is certified by two tangent lines; for non-symmetric laws the same structure gives the lower bound (Theorem 10.1) and is conjectured to be optimal:

setting	the two points	certificate
fixed moduli	$\pm \sum \arccos r_j$ (common sign)	concavity of $\log \cos$
symmetric laws	$\pm t \sum \mathbb{E} X_j $ (sign, mixing)	a tent at $\pm x_0$ (Lemma 4.8)
general laws (lower bound)	means of the two quantile parts	two tangents (Lemma 4.6)
scales, unimodal class	two scales	majorant of sinc (Lemma 8.1)
large d	odd multiples of π	odd-lattice threshold (Lemma 7.4)
unimodal moment problem	two atoms of Khinchine's Z	two-point decomposition (Lemma 11.2)

The mechanism is the same throughout: a single linear constraint survives, extreme points are two-point, and equality in the certificate forces the coupling to be a mix of the two parts — which is where complete and joint mixability (Wang–Wang, Puccetti–Wang–Wang) enter.

Organization. Section 2 collects tools; Section 3 recalls the free-law results of [1]; Section 4 develops the two-point principle; Sections 5–9 treat fixed marginals; Section 10 records the non-symmetric phenomenon; Section 11 identifies the extremal unimodal marginal without symmetry; Section 12 lists open problems. The finite verifications are in Appendix A.

2. PRELIMINARIES

Definition 2.1. For laws $F_1, \dots, F_d$ with characteristic functions φ_j ,

$$K(F_1, \dots, F_d) = \sup \left| \mathbb{E} e^{i \sum_j t_j X_j} - \prod_j \varphi_j(t_j) \right|$$

over $t \in \mathbb{R}^d$ and couplings with $X_j \sim F_j$; $K_d^F = K(F, \dots, F)$. For symmetric F with $0 < m = \mathbb{E}|X| < \infty$, $\kappa_d^F = \sup_{x>0} \{\varphi(x/(dm))^d - \cos x\}$.

For symmetric F we write $s_0 \in (0, \infty]$ for the first positive zero of φ ($s_0 = \infty$ if there is none; first lobe $[0, s_0)$), $s^* = \sup_{s \geq s_0} |\varphi(s)|$, $\gamma = f(0)m$ when F has a density f , and $x_c = 2.3311\dots$ for the root of $\cos x + x \sin x = 1$ in $(\frac{\pi}{2}, \pi)$.

Convex order and mixability. $Y \leq_{\text{cx}} V$ means $\mathbb{E}g(Y) \leq \mathbb{E}g(V)$ for all convex g . A d -tuple $(F_1, \dots, F_d)$ is *jointly mixable* (JM) if some coupling has $\sum_j X_j$ constant; F is *d-completely mixable* (d -CM) if $(F, \dots, F)$ is JM. We use three criteria:

- (M1) (Wang–Wang 2011 [2]) A law with a nonincreasing density on $[0, b]$ is d -CM iff its mean is at least b/d .
- (M2) (Wang–Wang 2016 [6], Thms 2.1(a) and 3.2) Laws with nonincreasing densities on $[0, l_j]$ and means μ_j are JM iff $\sum_j \mu_j \geq \max_j l_j$.
- (M3) (Puccetti–Wang–Wang 2012 [3], Thm 4.2) A law with a concave density on a bounded interval is d -CM for every $d \geq 3$.

Unimodality. By Khinchine's theorem (see [7]) a symmetric unimodal X is CW with $W \sim U[-1, 1]$ independent of $C \geq 0$, so $\varphi(s) = \mathbb{E} \text{sinc}(sC)$; if X has a density, $|X|$ has a nonincreasing density on $[0, b]$, $b = \text{ess sup } C$, and $m = \mathbb{E}C/2 \leq b/2$.

Lemma 2.2 (the first lobe). *sinc $z \geq 1 - z/\pi$ for all $z \geq 0$. Hence for symmetric unimodal F , $\varphi(s) \geq 1 - 2sm/\pi$; for $d \geq 3$ the interval $[0, \pi/(dm)]$ lies in the first lobe, $\varphi(\pi/(dm)) \geq 1 - 2/d > 0$, and $\kappa_d^F \geq 1 + P_d(\pi) > 1$, where $P_d(x) = \varphi(x/(dm))^d$.*

Proof. On $[0, \pi]$ let $q(z) = \sin z - z(\pi - z)/\pi$. Then $q(\pi - z) = q(z)$, $q(0) = q'(\pi) = 0$ and $q'(\pi/2) = 0$, and $q'' = \frac{2}{\pi} - \sin z$ is positive on $[0, z_1]$ and negative on $(z_1, \pi/2]$, $z_1 = \arcsin \frac{2}{\pi}$. On $[0, z_1]$, q is convex with a double zero at 0, so $q \geq 0$; on $[z_1, \pi/2]$, q is concave, so $q \geq \min\{q(z_1), q(\pi/2)\} \geq 0$ ($q(\pi/2) = 1 - \pi/4$). By symmetry $q \geq 0$ on $[0, \pi]$. On $[\pi, 2\pi]$, $\sin z \geq -(z - \pi) \geq -z(z - \pi)/\pi$. For $z \geq 2\pi$, $1 - z/\pi \leq -1 \leq \text{sinc } z$. Then $\varphi(s) = \mathbb{E} \text{sinc}(sC) \geq 1 - s\mathbb{E}C/\pi$, and $\mathbb{E}C = 2m$. $\square$

Lemma 2.3 (lattice separation). *For all real y : $1 - \cos y \geq \frac{2}{\pi^2} \text{dist}(y, 2\pi\mathbb{Z})^2$ and $1 + \cos y \geq \frac{2}{\pi^2} \text{dist}(y, \pi(2\mathbb{Z} + 1))^2$.*

Proof. With δ the distance, $|\delta| \leq \pi$ and $1 - \cos \delta = 2 \sin^2(\delta/2) \geq 2(\delta/\pi)^2$, since $\sin u \geq 2u/\pi$ on $[0, \pi/2]$; and $1 + \cos y = 1 - \cos(y - \pi)$. $\square$

3. FREE LAWS: THE PROFILE AND THE CONSTANT C_d

For complex random variables with $|Z_j| \leq 1$ let $\Delta = \mathbb{E} \prod_j Z_j - \prod_j \mathbb{E} Z_j$. Put $r_j = |\mathbb{E} Z_j|$, $\theta_j = \arccos r_j \in [0, \pi/2]$, $R = \prod_j r_j$, $s = \sum_j \theta_j$, and

$$C_d(r_1, \dots, r_d) = \sup\{|\Delta| : |Z_j| \leq 1, |\mathbb{E} Z_j| = r_j\}.$$

The following results are proved in the companion paper [1]; we recall them because the regimes with fixed marginals below are measured against them.

Theorem 3.1 (profile [1, Thm 2.1]). *For every $d \geq 1$ and $r \in [0, 1]^d$: $C_d(r_1, \dots, r_d) = R - \cos(\min\{s, \pi\})$. For $s \leq \pi$ it is attained by $Z_j = e^{i\varepsilon\theta_j}$ with a common fair sign ε ; for $s \geq \pi$ (value $1 + R$) by a law with $\prod_j Z_j \equiv -1$.*

Corollary 3.2 (the constant C_d [1, Cor. 2.2, Thm 2.4]). *The supremum of $|\mathbb{E} \prod Z_j - \prod \mathbb{E} Z_j|$ over all $|Z_j| \leq 1$ is*

$$C_d = \max_{0 \leq x \leq \pi} \{\cos^d(x/d) - \cos x\}; \quad C_2 = 1, \quad C_3 = 2/\sqrt{3}, \quad C_4 = 9/7,$$

and for $d \geq 3$ it is attained only when all $\arccos |\mathbb{E} Z_j|$ equal x_d^/d , where $x_d^* \in (0, \pi)$ is the unique maximizer.*

Proof. Only the last assertion needs a comment: by Theorem 3.1 and strict concavity of $\log \cos$ on $[0, \pi/2)$, $\prod \cos \theta_j \leq \cos^d(s/d)$ with equality iff the θ_j are equal, and the maximizer $x_d^* = d\theta_d^*$ is unique and interior by [1, Thm 2.4]. $\square$

4. THE TWO-POINT PRINCIPLE

Throughout this section $X_1, \dots, X_d \sim F$, $t \in \mathbb{R}^d$, $S = \sum_j t_j X_j$ and $z = \prod_j \varphi(t_j)$. The extremal couplings in every problem solved exactly in this paper make the phase sum two-point; this section collects the three facts behind that: a lower bound by two-point fusions, a dual certificate made of two tangent lines, and the fact that equality forces complete mixability.

4.1. Relaxation. Let U be uniform on $(0, 1)$ and g_j the quantile function of $\text{sgn}(t_j)X$. The comonotone sum is $V = \sum_j |t_j| g_j(U) = h(U)$, with h nondecreasing. For symmetric F , $g_j = F^{-1}$ and $V = AX$ with $A = \sum_j |t_j|$.

Lemma 4.1 (comonotone domination). *For every coupling, $S \leq_{\text{cx}} V$. Consequently $|\mathbb{E} e^{iS} - z| \leq R(V, z) := \sup\{|\mathbb{E} e^{iT} - z| : T \leq_{\text{cx}} V\}$.*

Proof. Standard (Meilijson–Nádas [4]; Dhaene et al. [5]): $\mathbb{E}(S - k)_+ \leq \mathbb{E}(V - k)_+$ for all k and $\mathbb{E}S = \mathbb{E}V$. $\square$

4.2. Quantile-split fusions and the lower bound.

Definition 4.2. For $p \in (0, 1)$ let $a_p = \mathbb{E}[V \mid U < p]$, $b_p = \mathbb{E}[V \mid U \geq p]$, $v_p = h(p)$. The fusion T_p equals a_p on $\{U < p\}$ and b_p on $\{U \geq p\}$; it satisfies $T_p \leq_{\text{cx}} V$, and its characteristic value is $w_p = p e^{ia_p} + (1-p) e^{ib_p}$.

For a law F and $p \in (0, 1)$ let $F_-^{(p)}$ and $F_+^{(p)}$ be the laws of $X_-^{(p)} = F^{-1}(pU)$ and $X_+^{(p)} = F^{-1}(p + (1-p)U)$ (the lower and upper p -parts).

Theorem 4.3 (fusions are attained under mixability of the parts). *Let $t_1 = \dots = t_d = t > 0$. If $F_-^{(p)}$ and $F_+^{(p)}$ are d -completely mixable, there is a coupling with $S = T_p$ (where $V = dtX$); hence $K_d^F \geq |w_p - \varphi(t)^d|$.*

Proof. Take an event E with $\mathbb{P}(E) = p$, independent of two complete mixes $(L_1, \dots, L_d)$ of $F_-^{(p)}$ and $(R_1, \dots, R_d)$ of $F_+^{(p)}$, and put $X_j = L_j$ on E and $X_j = R_j$ off E . Then $X_j \sim pF_-^{(p)} + (1-p)F_+^{(p)} = F$, and $t \sum_j X_j$ equals $dt \mathbb{E}L_1 = a_p$ on E and b_p off E . $\square$

Corollary 4.4 (symmetric laws). *Let F be symmetric and $|X|$ be d -completely mixable. Then $K_d^F \geq \varphi(t)^d - \cos(dt \mathbb{E}|X|)$ for every t ; hence $K_d^F \geq \kappa_d^F$. By Wang–Wang, this holds if $|X|$ has a nonincreasing density on $[0, b]$ and $d \geq b/\mathbb{E}|X|$.*

Proof. For $p = \frac{1}{2}$ the parts are the laws of $-|X|$ and $|X|$, so $a_{1/2} = -dt \mathbb{E}|X|$, $b_{1/2} = dt \mathbb{E}|X|$ and $w_{1/2} = \cos(dt \mathbb{E}|X|)$. (Equivalently: a common random sign times a complete mix of $|X|$.) $\square$

Remark 4.5. If F has a concave density on a bounded interval, both parts have concave densities for every p , hence are d -CM for $d \geq 3$ (Puccetti–Wang–Wang, Thm 4.2): every fusion is attained. This is the entry point of Sections 10 and 11.

4.3. Two tangents.

Lemma 4.6 (two-tangent certificate). *Let $f \in C^1(\mathbb{R})$, $p \in (0, 1)$, and let ℓ_a, ℓ_b be the tangent lines to f at $a = a_p$ and $b = b_p$, with $\ell'_a \leq \ell'_b$ and $\ell_a(v_p) = \ell_b(v_p)$. If $G = \max(\ell_a, \ell_b) \geq f$ on the convex hull of the range of V , then $\sup_{T \leq_{\text{cx}} V} \mathbb{E}f(T) = pf(a) + (1-p)f(b)$, attained by T_p .*

Proof. $T \leq_{\text{cx}} V$ takes values in that convex hull and G is convex, so $\mathbb{E}f(T) \leq \mathbb{E}G(T) \leq \mathbb{E}G(V)$. The lines cross at v_p and $\ell'_a \leq \ell'_b$, so $G(V) = \ell_a(V)$ on $\{U < p\}$ (where $V \leq v_p$) and $G(V) = \ell_b(V)$ on $\{U \geq p\}$. By linearity $\mathbb{E}G(V) = p\ell_a(a) + (1-p)\ell_b(b) = pf(a) + (1-p)f(b) = \mathbb{E}f(T_p)$. $\square$

Lemma 4.7 (stationarity). *If h is continuous at p , then $\frac{d}{dp}[pf(a_p) + (1-p)f(b_p)] = \ell_a(v_p) - \ell_b(v_p)$. In particular, at an interior maximizer in p the tangent lines cross at the split point v_p , as required in Lemma 4.6; the slope condition $\ell'_a \leq \ell'_b$ and the majorization must still be checked.*

Proof. $\frac{da_p}{dp} = \frac{v_p - a_p}{p}$ and $\frac{db_p}{dp} = \frac{b_p - v_p}{1-p}$, so the derivative is $f(a) - f(b) + f'(a)(v_p - a) + f'(b)(b - v_p) = \ell_a(v_p) - \ell_b(v_p)$. $\square$

4.4. The symmetric case: the window. For symmetric F and equal frequencies the fusion at $p = \frac{1}{2}$ is $\pm x_0$ with $x_0 = dt \mathbb{E}|X|$, and in the direction of -1 Lemma 4.6 becomes the classical “tent”. Let $x_c \approx 2.3311$ be the root in $(\frac{\pi}{2}, \pi)$ of $\cos x + x \sin x = 1$.

Lemma 4.8 (window). *For $x_0 \in [x_c, \pi]$ and all real s : $\cos s \geq \cos x_0 - \sin x_0(|s| - x_0)$, with equality only at $|s| = x_0$ if $x_c < x_0 < \pi$.*

Proof. By symmetry let $s \geq 0$ and put $L(s) = \cos x_0 - \sin x_0(s - x_0)$, $g = \cos - L$; then $g(x_0) = g'(x_0) = 0$ and $g'' = -\cos$. On $[\frac{\pi}{2}, \frac{3\pi}{2}]$, g is convex and tangent to 0 at x_0 , so $g \geq 0$. On $[0, \frac{\pi}{2}]$, g is concave, so its minimum is at an end: $g(\frac{\pi}{2}) \geq 0$, and $g(0) = 1 - \cos x_0 - x_0 \sin x_0 \geq 0$ because $x \mapsto \cos x + x \sin x$ decreases on $[\frac{\pi}{2}, \pi]$ and equals 1 at x_c . For $s \geq \frac{3\pi}{2}$, $L(s) \leq L(\frac{3\pi}{2}) \leq -1$, since $1 + \cos x_0 \leq \sin x_0(\frac{3\pi}{2} - x_0)$ is $\cot \frac{x_0}{2} \leq \frac{3\pi}{2} - x_0$, true on $[x_c, \pi]$ ($\cot \frac{x_c}{2} < 0.43 < \frac{\pi}{2}$). Strictness away from x_0 follows from strict convexity or concavity on each piece when $x_c < x_0 < \pi$. $\square$

Corollary 4.9. *If F is symmetric, $x_0 = A \mathbb{E}|X| \in [x_c, \pi]$ and $T \leq_{\text{cx}} AX$, then $\mathbb{E} \cos T \geq \cos x_0$, attained by the fusion $\pm x_0$.*

Proof. $\mathbb{E} \cos T \geq \cos x_0 - \sin x_0(\mathbb{E}|T| - x_0) \geq \cos x_0$, since $\mathbb{E}|T| \leq \mathbb{E}|AX| = x_0$. $\square$

4.5. Equality forces complete mixability.

Proposition 4.10. *Let $t_1 = \dots = t_d = t > 0$, $V = dtX$, and suppose a coupling gives $S = T_p$ almost surely, i.e. S equals a_p on an event E with $\mathbb{P}(E) = p$ and b_p on E^c . If F is continuous at $F^{-1}(p)$, then on E every X_j has law $F_-^{(p)}$ and on E^c law $F_+^{(p)}$; in particular both parts are d -completely mixable.*

Proof. (Bathtub principle; $t > 0$.) For any event B with $\mathbb{P}(B) = p$, $\mathbb{E}[X; B] \geq \mathbb{E}[X; X < F^{-1}(p)] = p \mathbb{E}X_-^{(p)}$, with equality only if $B = \{X < F^{-1}(p)\}$ up to null sets. Summing over j : $\mathbb{E}[S; E] = t \sum_j \mathbb{E}[X_j; E] \geq p dt \mathbb{E}X_-^{(p)} = p a_p$, and $\mathbb{E}[S; E] = p a_p$ by assumption; so equality holds for every j . $\square$

Corollary 4.11 (symmetric case). *If F is symmetric, $t > 0$, $x_0 = dt \mathbb{E}|X| \in (x_c, \pi)$ and a coupling gives $\mathbb{E} \cos S = \cos x_0$ with $S = t \sum X_j$, then $\sum_j |X_j| \equiv d \mathbb{E}|X|$ almost surely.*

Proof. By Lemma 4.8 and Corollary 4.9, equality forces $\mathbb{E}|S| = x_0$ and $|S| = x_0$ a.s. Since $|S| \leq t \sum |X_j|$ and both sides have mean x_0 , $t \sum_j |X_j| \equiv x_0$. $\square$

5. SYMMETRIC UNIMODAL MARGINALS: THE MAIN THEOREM

Let F be symmetric and unimodal with density f , $m = \mathbb{E}|X|$, $\gamma = f(0)m$, and let $s_0 \in (0, \infty]$ be the first positive zero of φ (the *first lobe* is $[0, s_0]$), $s^* = \sup_{s \geq s_0} |\varphi(s)|$ ($s^* = 0$ if $s_0 = \infty$), and $P_d(x) = \varphi(x/(dm))^d$. For $0 < x_0 \leq \pi$ and $\eta \in [0, \pi]$ put $\hat{x}_0 = \max\{x_0, x_c\}$ and

$$B_d(x_0, \eta) = \min \left\{ 1 + P_d(x_0) \cos \eta, \quad -\cos \hat{x}_0 + \sin \hat{x}_0 (x_0 - \hat{x}_0 + \min\{\frac{\gamma \eta^2}{x_0}, \eta\}) + P_d(x_0) \cos \eta \right\}.$$

Conditions. $(W_d) \sup_{0 < x_0 \leq \pi, 0 \leq \eta \leq \pi} B_d(x_0, \eta) \leq \kappa_d^F$; $(T_d) 1 + s^* \leq \kappa_d^F$.

Theorem 5.1 (main theorem). *Let F be symmetric and unimodal with density, with $\log \varphi$ concave on its first lobe.*

- (i) *If (W_d) and (T_d) hold, then $K_d^F \leq \kappa_d^F$.*
- (ii) *If F is supported on $[-b, b]$ and $d \geq b/m$, then $K_d^F \geq \kappa_d^F$.*
- (iii) *If F has bounded support and $\gamma < \gamma_0 := \min_{x_c \leq x \leq \pi} \frac{2x}{\pi^2 \sin x} = 0.6519\dots$, then there is d_F with $K_d^F = \kappa_d^F$ for all $d \geq d_F$.*

Remark 5.2. In the window $x_0 \geq x_c$, if $\gamma \sin x_0 \eta^2 / x_0 \leq P_d(x_0)(1 - \cos \eta)$ for all η , the second term of B_d is at most $P_d(x_0) - \cos x_0 \leq \kappa_d^F$: this is the two-tangent certificate of Section 4 in every direction (“exactness”). The finite verifications of the Appendix check (W_d) by accepting each box either by a margin or by exactness.

Lemma 5.3 (shift of a unimodal law). *If V is symmetric and unimodal with density g , then $\mathbb{E}|V - \eta| \leq \mathbb{E}|V| + \min\{g(0)\eta^2, |\eta|\}$ for all η .*

Proof. $\mathbb{E}|V - \eta| - \mathbb{E}|V| = \int_0^{|\eta|} (2F_V(u) - 1) du$, and $0 \leq 2F_V(u) - 1 \leq \min\{2g(0)u, 1\}$ for $u \geq 0$ because $g \leq g(0)$. $\square$

Lemma 5.4 (clamped tent). *Let V be symmetric and unimodal with density g , $x_0 = \mathbb{E}|V| \in (0, \pi]$, $\gamma_V = g(0)x_0$, and $S' \leq_{\text{cx}} V$. Then for every real η ,*

$$\mathbb{E} \cos(S' - \eta) \geq \cos \hat{x}_0 - \sin \hat{x}_0 (x_0 - \hat{x}_0 + \min\{\gamma_V \eta^2 / x_0, |\eta|\}).$$

For $V = TX$ with $T > 0$, $\gamma_V = \gamma$.

Proof. Lemma 4.8 at $\hat{x}_0 \in [x_c, \pi]$, applied pointwise at $s = S' - \eta$, gives $\cos(S' - \eta) \geq \cos \hat{x}_0 - \sin \hat{x}_0 (|S' - \eta| - \hat{x}_0)$. Take expectations, use $\mathbb{E}|S' - \eta| \leq \mathbb{E}|V - \eta|$ (convex order) and Lemma 5.3. For $V = TX$, $g(0) = f(0)/T$ and $x_0 = Tm$. $\square$

Proof of Theorem 5.1. (i) Reductions. Replacing X_j by $-X_j$ does not change the law, so let $t_j \geq 0$; put $T = \sum_j t_j$, $x_0 = Tm$ and $P = \prod_j \varphi(t_j)$ (real). By Lemma 4.1, $|\mathbb{E}e^{iS} - P| \leq R(T, P) := \sup\{|w - P| : w \in W_T\}$, $W_T = \{\mathbb{E}e^{iS'} : S' \leq_{\text{cx}} TX\}$; the map $P \mapsto R(T, P)$ is convex, $R(T, 0) \leq 1$, and W_T is invariant under conjugation.

Tail. If some $t_j \geq s_0$, then $|P| \leq s^*$ and the defect is at most $1 + s^* \leq \kappa_d^F$ by (T_d).

First lobe. Otherwise all $t_j < s_0$ and $0 < P \leq P_d(x_0)$ by concavity of $\log \varphi$. By convexity, $R(T, P) \leq \max\{R(T, 0), R(T, P_d(x_0))\} \leq \max\{1, R(T, P_d(x_0))\}$, and $1 < \kappa_d^F$ (Lemma 2.2); so it suffices to bound $R(T, P_d(x_0))$.

$x_0 > \pi$. Since $\log \varphi$ is concave with $(\log \varphi)'(0) = 0$, φ decreases on the first lobe, which contains $[\pi/(dm), x_0/(dm)]$ (Lemma 2.2); so $R \leq 1 + P_d(x_0) \leq 1 + P_d(\pi)$, the value of the formula at $x = \pi$, hence $\leq \kappa_d^F$.

$0 < x_0 \leq \pi$. For $w \in W_T$, $|w - P| = \sup_\eta \{-\operatorname{Re}(e^{-i\eta}w) + P \cos \eta\}$ (direction $\pi + \eta$), and by conjugation invariance $\eta \in [0, \pi]$ suffices. Now $-\operatorname{Re}(e^{-i\eta}w) = -\mathbb{E} \cos(S' - \eta)$ is at most 1 and, by Lemma 5.4, at most $-\cos \hat{x}_0 + \sin \hat{x}_0(x_0 - \hat{x}_0 + \min\{\gamma\eta^2/x_0, \eta\})$. Hence $R(T, P_d(x_0)) \leq \sup_\eta B_d(x_0, \eta) \leq \kappa_d^F$ by (W_d).

(ii) $|X|$ has a nonincreasing density on $[0, b]$ with mean $m \geq b/d$, so it is d -completely mixable (Wang–Wang); apply Corollary 4.4.

(iii) Let $b < \infty$ bound the support. For $d \geq b/m$, (ii) applies. As $d \rightarrow \infty$, $P_d \rightarrow 1$ uniformly on $[0, \pi]$ and $\kappa_d^F \geq 1 + P_d(\pi) \rightarrow 2$, so (T_d) holds since $s^* < 1$. In (W_d) with $x_0 \geq x_c$: since $(1 - \cos \eta)/\eta^2 \geq 2/\pi^2$ on $(0, \pi]$ and $\gamma < \gamma_0$, we get $\gamma \sin x_0 \eta^2/x_0 \leq (1 - \delta)(1 - \cos \eta)$ for some $\delta > 0$, hence exactness once $P_d \geq 1 - \delta$. For $x_0 < x_c$ (where $\hat{x}_0 = x_c$ and the second term is $-1 + \sin x_c(x_0 + \min\{\gamma\eta^2/x_0, \eta\}) + P_d \cos \eta$, since $\cos x_c + x_c \sin x_c = 1$): if $\eta \geq \frac{1}{2}$, the first term is at most $1 + \cos \frac{1}{2} < 1.88$; if $\eta \leq \frac{1}{2}$, then $x_0 + \min\{\gamma\eta^2/x_0, \eta\} \leq \max\{(1 + \gamma)\eta, x_c + \gamma\eta^2/x_c\} \leq 2.41$, so the second term is at most $-1 + 2.41 \sin x_c + 1 < 1.75$. Hence $\sup B_d < 1.88$ there, while $\kappa_d^F \rightarrow 2$. $\square$

Lemma 5.5 (one frequency beyond the lobe). *Let F be symmetric, $t_1 \geq 0$ arbitrary and $t_2, \dots, t_d$ in the first lobe, $R' = \prod_{j \geq 2} \varphi(t_j)$, $P = \varphi(t_1)R'$ and $\beta = m \sum_{j \geq 2} t_j \leq \pi$. Then for every coupling $|\mathbb{E}e^{iS} - P| \leq |\varphi(t_1)| - |P| + 2 \sin(\beta/2)$.*

Proof. Let $Z_1 = e^{it_1 X_1}$, $Y = \sum_{j \geq 2} t_j X_j$, $W = e^{iY}$. Then $\mathbb{E}e^{iS} - P = \varphi(t_1)(1 - R') + \mathbb{E}[Z_1(W - 1)]$, and $0 < R' \leq 1$. Since $|W - 1| = 2|\sin(Y/2)| \leq 2g(|Y|)$ with the concave nondecreasing $g(y) = \sin(y/2)$ for $y \leq \pi$, $g = 1$ beyond, Jensen gives $\mathbb{E}|W - 1| \leq 2g(\mathbb{E}|Y|) \leq 2 \sin(\beta/2)$, as $\mathbb{E}|Y| \leq \beta$. $\square$

When (T_d) fails, this lemma replaces it: in the region $|P| > \kappa_d^F - 1$, log-concavity bounds $R' \leq \varphi(\beta/((d-1)m))^{d-1}$, which caps β , while the lemma settles small β (Theorem 8.7 and Lemma A.3).

Theorem 5.6 (different laws). *Let $F_1, \dots, F_d$ be symmetric unimodal with densities f_j , $0 < m_j = \mathbb{E}|X_j| < \infty$, $b_j = \operatorname{ess\,sup} |X_j|$, $\log \varphi_j$ concave on the first lobes, $\gamma = \max_j f_j(0)m_j$, $s^* = \max_j s_j^*$, and*

$$K^* = \sup_{t \geq 0} \left\{ \prod_j \varphi_j(t_j) - \cos \sum_j t_j m_j \right\}.$$

(i) *If $1 + s^* \leq K^*$ and $B(x_0, \eta; P(t)) \leq K^*$ for all t in the first lobes with $x_0 = \sum t_j m_j \leq \pi$ and all $\eta \in [0, \pi]$ (with B as B_d with $P_d(x_0)$ replaced by $P(t) = \prod \varphi_j(t_j)$), then $K(F_1, \dots, F_d) \leq K^*$.*
(ii) *If K^* is attained at t^* with $\sum_j t_j^* m_j \geq \max_j t_j^* b_j$, then $K(F_1, \dots, F_d) \geq K^*$.*

Proof. (i) By symmetry of each F_j we may take $t \geq 0$. As in Theorem 5.1, now with $V = \sum_j t_j F_j^{-1}(U)$. The quantile function of V is a sum of quantile functions that are convex on $(\frac{1}{2}, 1)$ (for a symmetric unimodal density $q'(u) = 1/f(q(u))$ increases there), so V is symmetric and unimodal, with density $g(0) = 1/\sum_j t_j/f_j(0)$ at 0 and $g(0)\mathbb{E}|V| = \sum t_j m_j / \sum (t_j/f_j(0)) \leq \gamma$ (mediant inequality); so Lemma 5.4 holds with $\gamma_V \leq \gamma$, and B increases in γ . No reduction of unequal frequencies is needed, because K^* is a supremum over all t . If $x_0 > \pi$, scaling t to $x_0 = \pi$ increases every $\varphi_j(t_j)$ on the first lobes, so $1 + P(t) \leq K^*$. (ii) The moduli $t_j^*|X_j|$ have nonincreasing densities on $[0, t_j^* b_j]$ with means $t_j^* m_j$; by (M2) they are JM, and a common random sign gives $S = \pm x_0^*$, hence the defect $P(t^*) - \cos x_0^* = K^*$. $\square$

6. FAMILIES AND A TABLE OF CONSTANTS

Corollary 6.1. $K_d^F = \kappa_d^F$ for: the uniform law and the triangular law, all $d \geq 3$; the cubic B-spline (sum of three $U[-\frac{1}{2}, \frac{1}{2}]$) and the Epanechnikov law, all $d \geq 4$ (and $K_3^F \leq \kappa_3^F$ for the B-spline). For the uniform law on $[-1, 1]$, $\kappa_d = \max_x \{\text{sinc}(2x/d)^d - \cos x\}$; $\kappa_3 = 1.0849633414$, $\kappa_4 = 1.1907222653$.

Proof. Theorem 5.1(i)–(ii), with the hypotheses verified in Appendix A (log-concavity, tail, (W_d)); where (T_d) fails (uniform, $d = 3, 4$) the tail is closed by Lemma 5.5 and Lemma A.3 for one frequency beyond the lobe, and by $|P| \leq s^{*2} = 0.0472 < \kappa_d - 1$ for two or more. Mixability: $b/m = 2$ (uniform), 3 (triangular), $48/13$ (B-spline), $8/3$ (Epanechnikov). $\square$

law	$d = 3$	$d = 4$	status
free laws, C_d	$1.1547005384 = 2/\sqrt{3}$	$1.2857142857 = 9/7$	sharp
uniform $[-1, 1]$	1.0849633414	1.1907222653	$K_d = \kappa_d$, all $d \geq 3$
triangular	1.0765878231	1.1642014101	$K_d = \kappa_d$, all $d \geq 3$
cubic B-spline	1.0808671327	1.1656390998	$K_3 \leq \kappa_3$; $K_d = \kappa_d$, $d \geq 4$
Epanechnikov	1.0800186407	1.1758685660	$K_d = \kappa_d$, $d \geq 4$ ($d = 3$ open)
$\frac{3}{4}U[-5, 5] + \frac{1}{4}U[-9, 9]$	1.0874848676		$K_3 = \kappa_3^F > \kappa_3$
Gaussian	1.0848101338	1.1630296791	$K_d < \kappa_d^G$ (bound not attained)

TABLE 1. Values of $\kappa_d^F = \max_x \{\varphi(x/(d\mathbb{E}|X|))^d - \cos x\}$ (and C_d), computed to 30 digits.

7. THE PRICE OF FIXING THE MARGINALS

Let X have support in $[a, b]$, $\mu = \mathbb{E}X$, m a median, $\sigma^2 = \text{Var}X > 0$, $\tau = \mathbb{E}|X - m|$.

Theorem 7.1. If F is lattice or satisfies Cramér's condition ($\sup_{|t| \geq T} |\varphi(t)| < 1$ for every $T > 0$), then

$$K_d^F = 2 - \frac{\pi^2 \sigma^2}{2d \tau^2} + O(d^{-3/2}) \quad (d \rightarrow \infty).$$

No symmetry, unimodality or mixability is assumed: the median replaces the centre of symmetry. Since $\sigma \geq \mathbb{E}|X - \mu| \geq \tau$, the factor $\sigma^2/\tau^2 \geq 1$ is the *price of fixing the marginals*, with equality iff X is two-point with probabilities $\frac{1}{2}$ (the rate of C_d).

Corollary 7.2. For symmetric F (bounded support, lattice or Cramér), $2 - K_d^F = \frac{\pi^2}{2d} \frac{\text{Var}X}{(\mathbb{E}|X|)^2} + O(d^{-3/2})$; for the uniform law the factor is $4/3$, for the triangular law $3/2$.

Lemma 7.3 (odd lattice). For integrable Y and $|c| \leq \pi$: $\mathbb{E}|Y - c| \geq \pi - |\mathbb{E}Y| - 2\mathbb{E} \text{dist}(Y, \pi(2\mathbb{Z} + 1))$.

Proof. Let Y' be a nearest odd multiple of π , $D = |Y - Y'|$, $\nu = \mathbb{E}Y'$; then $|\nu| \leq |\mathbb{E}Y| + \mathbb{E}D$ and $\mathbb{E}|Y - c| \geq \mathbb{E}|Y' - c| - \mathbb{E}D$. From $|y - c| \geq |y| - c \text{sgn } y$ and, if $P_+ = \mathbb{P}(Y' > 0) \geq \frac{1}{2}$, $\mathbb{E}|Y'| = 2\mathbb{E}Y'_+ - \nu \geq 2\pi P_+ - \nu$, we get $\mathbb{E}|Y' - c| \geq P_+(2\pi - 2c) + c - \nu \geq \pi - \nu$; the case $P_- \geq \frac{1}{2}$ is symmetric. $\square$

Lemma 7.4 (threshold). Let $V = \sum_j s_j g_j(U)$ with $s_j \geq 0$, $A = \sum s_j$, each g_j the quantile function of X or $-X$. If $\mathbb{E}Y = 0$ and $Y \leq_{\text{cx}} V - \mathbb{E}V$, then $\mathbb{E} \cos Y \geq -1 + \frac{\tau^2}{2\pi^2} (\pi/\tau - A)^2_+$.

Proof. All $g_j(U) - g_j(\frac{1}{2})$ have the sign of $U - \frac{1}{2}$, so $\text{med } V = \sum s_j g_j(\frac{1}{2})$ and $\mathbb{E}|V - \text{med } V| = A\tau$. With $c = \text{med } V - \mathbb{E}V$ we have $|c| \leq A\tau$ and, by convexity, $\mathbb{E}|Y - c| \leq A\tau$. If $A\tau < \pi$, Lemma 7.3 gives $\mathbb{E}D \geq (\pi - A\tau)/2$ for $D = \text{dist}(Y, \pi(2\mathbb{Z} + 1))$, and Lemma 2.3 gives $\mathbb{E}(1 + \cos Y) \geq \frac{2}{\pi^2} (\mathbb{E}D)^2$. $\square$

Proof of Theorem 7.1. Lattice reduction. If $X \in a + h\mathbb{Z}$ with maximal span h , replacing t_j by $t_j + 2\pi/h$ multiplies $\mathbb{E}e^{iS}$ and $z = \prod \varphi(t_j)$ by the same unimodular constant; so we may take $|t_j| \leq \pi/h$. In both cases there are $T_0 > 0$ and $q < 1$ with $|\varphi(t)| \leq q$ for $T_0 \leq |t|$ (in the reduced range), and near 0: $\log |\varphi(t)| \leq -\frac{\sigma^2}{2}t^2 + C_1 t^4$, $|\varphi(t)| \leq e^{-\sigma^2 t^2/4}$, $|\arg \varphi(t) - \mu t| \leq C_2 |t|^3$ ($|\varphi|^2$ is even).

Upper bound. Let $\Delta = |\mathbb{E}e^{iS} - z|$. (i) If some $|t_j| \geq T_0$, $\Delta \leq 1 + q$. (ii) If all $|t_j| < T_0$ and $\sum t_j^2 \geq M/d$ with $M > 4\pi^2/\tau^2$, $\Delta \leq 1 + e^{-\sigma^2 M/(4d)} \leq 2 - \sigma^2 M/(8d)$. (iii) Otherwise $A = \sum |t_j| \leq \sqrt{M}$; write $\mathbb{E}e^{iS} = e^{i\mathbb{E}S}u$ with $u = \mathbb{E}e^{iY}$, $Y = S - \mathbb{E}S \leq_{\text{cx}} V - \mathbb{E}V$ (Lemma 4.1), and $z = \rho e^{i(\mathbb{E}S + \varepsilon)}$, $|\varepsilon| \leq C_2 \sum |t_j|^3 = O(d^{-3/2})$. By Lemma 7.4, $\text{Re } u \geq -1 + \eta$ with $\eta = \kappa(\pi/\tau - A)_+^2$, $\kappa = \tau^2/(2\pi^2)$, so $|u - \rho|^2 \leq (1 + \rho)^2 - 2\rho\eta$ and $\Delta \leq 1 + \rho - \frac{\rho\eta}{1+\rho} + \rho|\varepsilon|$. Also $\rho \leq \exp(-\sigma^2 A^2/(2d) + O(d^{-2}))$ since $\sum t_j^2 \geq A^2/d$. With $A^* = \pi/\tau$, $x = (A^* - A)_+$ and $\rho \geq \frac{1}{2}$ (otherwise $\Delta \leq \frac{3}{2}$),

$$\Delta \leq 1 + e^{-\sigma^2 A^{*2}/(2d)} \left(1 + \frac{2\sigma^2 A^* x}{d}\right) - \frac{\kappa}{3} x^2 + O(d^{-3/2}) \leq 2 - \frac{\sigma^2 A^{*2}}{2d} + O(d^{-3/2}),$$

since $\max_{x \geq 0} \left\{ \frac{2\sigma^2 A^*}{d} x - \frac{\kappa}{3} x^2 \right\} = \frac{3\sigma^4 A^{*2}}{\kappa d^2} = O(d^{-2})$ and $e^{-y} \leq 1 - y + y^2/2$.

Lower bound (a fusion with approximate mixing, cf. Section 4). Split at the median: $F_{\mp}$ are the laws of $F^{-1}(U/2)$ and $F^{-1}((1+U)/2)$, with means $\mu_{\mp}$; then $\mu_+ - \mu_- = 2\tau$ and $\mu - \mu_- = \tau$. On an event of probability $\frac{1}{2}$, independent of the uniform variables U, U' , put $X_j = F_-^{-1}(\{U + j/d\})$, otherwise $X_j = F_+^{-1}(\{U' + j/d\})$: each $X_j \sim F$, and since a monotone function's Riemann sums at shifted grids are within its range of the integral, the two sums are $d\mu_{\mp} + e_{\mp}$ with $|e_{\mp}| \leq b - a$. At $t = \pi/(d\tau)$, $e^{itd\mu_+} = e^{itd\mu_-}$ and $\mathbb{E}e^{iS} = e^{itd\mu_-} v$ with $\text{Re } v \geq 1 - t^2(b-a)^2/2$, while $z = -\rho e^{itd\mu_-} e^{i\beta}$ with $\beta = O(d^{-2})$ and $\rho = \exp(-\pi^2 \sigma^2/(2d\tau^2) + O(d^{-3}))$. Hence $\Delta \geq \text{Re}(v + \rho e^{i\beta}) \geq 1 + \rho - O(d^{-2})$. $\square$

8. EXTREMALITY OF THE UNIFORM LAW

By Khinchine's theorem a symmetric unimodal X is CW with $W \sim U[-1, 1]$ independent of $C \geq 0$, so $\varphi(t) = \mathbb{E} \text{sinc}(tC)$, $m = \mathbb{E}C/2$ and $b = \text{ess sup } C$; the uniform law with the same m is $U[-2m, 2m]$. With $Y = C/\mathbb{E}C$: $\varphi(x/(dm)) = \mathbb{E} \text{sinc}(zY)$, $z = 2x/d$, and $b \leq dm$ means $Y \leq d/2$.

Lemma 8.1 (majorant of sinc). *Let $\hat{h} = \text{sinc}$ on $[0, \frac{\pi}{2}]$ and $\hat{h}(z) = \frac{4}{\pi^2}(\pi - z)$ on $[\frac{\pi}{2}, \pi]$. Then $\hat{h}$ is concave on $[0, \pi]$ and $\hat{h} \geq \text{sinc}$.*

Proof. The line is the tangent to sinc at $\frac{\pi}{2}$ ($\text{sinc } \frac{\pi}{2} = \frac{2}{\pi}$, $\text{sinc}' \frac{\pi}{2} = -\frac{4}{\pi^2}$) and sinc is concave on $[0, \frac{\pi}{2}]$, so $\hat{h}$ is C^1 and concave. On $[\frac{\pi}{2}, \pi]$, $\hat{h} \geq \text{sinc}$ is $p(z) = \frac{4}{\pi^2}z(\pi - z) - \sin z \geq 0$. Now p is symmetric about $\frac{\pi}{2}$ with $p(0) = p(\frac{\pi}{2}) = 0 = p'(\frac{\pi}{2})$, and $p'' = -\frac{8}{\pi^2} + \sin z$ increases on $[0, \frac{\pi}{2}]$: p is concave then convex there, convex near $\frac{\pi}{2}$ with a double zero, hence $p \geq 0$. $\square$

Lemma 8.2 (dominance). *If $Y \geq 0$, $\mathbb{E}Y = 1$, $0 < z \leq \frac{\pi}{2}$ and $zY \leq \pi$, then $\mathbb{E} \text{sinc}(zY) \leq \text{sinc } z$, with equality only if $Y \equiv 1$.*

Proof. $\mathbb{E} \text{sinc}(zY) \leq \mathbb{E} \hat{h}(zY) \leq \hat{h}(z) = \text{sinc } z$ (Jensen). Equality in Jensen needs zY inside an interval where $\hat{h}$ is affine and containing z ; the only affine piece is $[\frac{\pi}{2}, \pi]$, which forces $z = \frac{\pi}{2}$ and $Y \geq 1$, hence $Y \equiv 1$. $\square$

Theorem 8.3. *Let $d \geq 4$ and F symmetric unimodal with $b \leq dm$, φ nonincreasing on its first lobe and $s^* \leq \text{sinc}(2\pi/d)$. Then $\kappa_d^F \leq \kappa_d$, with equality iff F is uniform. Hence, whenever $K_d^F = \kappa_d^F$ (Theorem 5.1), the uniform law has the largest constant among these symmetric unimodal marginals.*

Proof. For $0 < x \leq \pi$: $z = 2x/d \leq \pi/2$ because $d \geq 4$, and $zY \leq x \leq \pi$ because $Y \leq d/2$; so $0 \leq \varphi(x/(dm)) \leq \text{sinc}(2x/d)$ by Lemma 8.2 ($\text{sinc} \geq 0$ on $[0, \pi]$), and $\varphi(x/(dm))^d - \cos x \leq \text{sinc}(2x/d)^d - \cos x \leq \kappa_d$. For $x > \pi$: $|\varphi(x/(dm))| \leq \max\{\varphi(\pi/(dm)), s^*\} \leq \text{sinc}(2\pi/d)$, so the value is at most $1 + \text{sinc}(2\pi/d)^d$, the uniform formula at π , which is $< \kappa_d$ since its maximizer lies in $(0, \pi)$. Equality forces a maximizer in $(0, \pi)$ and $Y \equiv 1$ there. $\square$

Remark 8.4 (why $d \geq 4$). The majorant $\hat{h}$ touches sinc exactly up to $\frac{\pi}{2}$, and $2\pi/d \leq \frac{\pi}{2}$ iff $d \geq 4$. For $d = 3$ the relevant argument $2x^*/3 \approx 1.98$ lies beyond $\frac{\pi}{2}$, where mixing two scales helps.

Remark 8.5 (symmetry is needed). The symmetry assumption in Theorem 8.3 cannot be dropped. The law $F = \frac{1}{4}\delta_0 + \frac{3}{4}U[0, 1]$ is unimodal, and $\varphi(t) = \frac{1}{4} + \frac{3}{4}\frac{e^{it}-1}{it}$ satisfies $|\varphi(t)| < 1$ for $t \neq 0$ and $|\varphi(t)| \rightarrow \frac{1}{4}$, so Cramér's condition holds. Its median is $\frac{1}{3}$, $\mathbb{E}|X - \frac{1}{3}| = \frac{7}{24}$ and $\text{Var}X = \frac{7}{64}$, so $\text{Var}X/(\mathbb{E}|X - \text{med}X|)^2 = \frac{9}{7} < \frac{4}{3}$. By Theorem 7.1, $K_d^F > K_d^U$ for all sufficiently large d . (In fact $\frac{9}{7}$ is the smallest value of this ratio over unimodal laws, and $K_d^F > K_d^U$ already for every $d \geq 4$; see Section 11.)

Theorem 8.6 (different laws). *Let $d \geq 4$ and let $F_1, \dots, F_d$ satisfy Theorem 5.6, with the maximum attained at t^* where joint mixability holds. Then $K(F_1, \dots, F_d) \leq \kappa_d$, with equality iff all F_j are uniform.*

Proof. Scaling t down to $x_0 = \pi$ does not decrease the formula on the first lobes, so $x_0^* = \sum t_j^* m_j \leq \pi$. With $u_j = t_j^* m_j$ and $Y_j = C_j/\mathbb{E}C_j$, joint mixability gives $2u_j Y_j = t_j^* C_j \leq t_j^* b_j \leq x_0^* \leq \pi$ and $2u_j \leq t_j^* b_j$ (as $b_j \geq 2m_j$). Hence $\varphi_j(t_j^*) \leq \hat{h}(2u_j)$ (Jensen), and since $\hat{h}$ is concave and positive on $[0, \pi)$, $\log \hat{h}$ is concave: $\prod_j \hat{h}(2u_j) \leq \hat{h}(2x_0^*/d)^d = \text{sinc}(2x_0^*/d)^d$. So $K \leq \text{sinc}(2x_0^*/d)^d - \cos x_0^* \leq \kappa_d$.

Equality. $\log \hat{h}$ is strictly concave on $[0, \pi)$ ($\log \text{sinc}$ on $[0, \frac{\pi}{2}]$, $\log(\pi - z)$ on $[\frac{\pi}{2}, \pi)$), so $K = \kappa_d$ forces $u_j = x_0^*/d$ for all j . Then $2u_j = 2x_0^*/d \leq \frac{\pi}{2}$ and $\varphi_j(t_j^*) = \mathbb{E} \text{sinc}(2u_j Y_j) = \text{sinc}(2u_j)$, so Lemma 8.2 gives $Y_j \equiv 1$: each F_j is uniform. Conversely, for uniform F_j (of any scales) the formula reduces to $\max_x \{\text{sinc}(2x/d)^d - \cos x\} = \kappa_d$ at equal u_j , where the moduli are identical uniforms and hence jointly mixable. $\square$

Theorem 8.7 ($d = 3$: the threshold is sharp). *For $F^* = \frac{3}{4}U[-5, 5] + \frac{1}{4}U[-9, 9]$: $K_3^{F^*} = \kappa_3^{F^*} = 1.087484867576 \dots > \kappa_3 = 1.084963341374 \dots$*

Proof. Lower bound. $|X|$ has a nonincreasing density on $[0, 9]$ and mean $3 = 9/3$, so it is 3-CM (Wang–Wang); Corollary 4.4 at $x = 2.97$ gives the value. *Upper bound.* Here $m = 3$, $\gamma = 4/15$, $\log \varphi$ is concave on the first lobe $[0, 0.59949]$ (Appendix), and (W_3) holds (Appendix: 54 boxes). (T_3) fails ($s^* \leq 0.13537$), so we use Lemma 5.5: with $\tau = \kappa_3^{F^*} - 1$, the region $|P| > \tau$ needs $\varphi(\beta/6)^2 \geq R' > \tau/s^*$, i.e. $\beta < \beta_{\max} = 6\varphi^{-1}(\sqrt{\tau/s^*}) = 1.0891$, while $|\varphi(t_1)| - |P| + 2\sin(\beta/2) \leq s^* - \tau + 2\sin(\beta/2) \leq \kappa_3^{F^*}$ for $\beta \leq \beta_1 = 1.0932$. Two frequencies beyond the lobe give $|P| \leq s^{*2} < \tau$. $\square$

Proposition 8.8 (optimal κ -profile over the $d = 3$ unimodal class). *Over symmetric unimodal laws with 3-completely mixable $|X|$, $\sup_F \kappa_3^F = \sup_x \{\hat{h}_x(1)^3 - \cos x\} = 1.0874848690 \dots$, where $\hat{h}_x$ is the least concave majorant of $y \mapsto \text{sinc}(2xy/3)$ on $[0, \frac{3}{2}]$; it is attained by the normalized scale $Y = C/\mathbb{E}C \in \{0.83346, \frac{3}{2}\}$ with probabilities $0.75014, 0.24986$, and F^* is within $1.4 \cdot 10^{-9}$.*

This concerns the formula κ_3^F . Whether $\sup_F K_3^F$ over the same class equals this value is open (Section 12); it holds for F^* by Theorem 8.7.

Proof. For fixed x the constraint $\mathbb{E}Y = 1$ on $Y \in [0, \frac{3}{2}]$ is one linear constraint, so $\sup \mathbb{E} \text{sinc}(2xY/3)$ is attained by two-point laws and equals the concave majorant at 1 — the two-point principle once more. $\square$

9. ATTAINMENT, GAUSSIAN MARGINALS AND THE MIXABILITY CONJECTURE

Strict conditions. (W_d^s) : $B_d(x_0, \eta) < \kappa_d^F$ for all $(x_0, \eta) \neq (x_0^*, 0)$, where $x_0^* \in (x_c, \pi)$ is the unique maximizer of $P_d(x) - \cos x$; (T_d^s) : $1 + s^* < \kappa_d^F$.

Theorem 9.1 (attainment forces mixability). *Let F be symmetric unimodal with density, $\log \varphi$ strictly concave on its first lobe, and (W_d^s) , (T_d^s) . If $K_d^F = \kappa_d^F$, then some coupling has $\sum_j |X_j| \equiv dm$. Consequently, under these conditions, $K_d^F = \kappa_d^F$ if and only if $|X|$ is d -completely mixable.*

Proof. Take frequencies $t^{(n)} \geq 0$ and couplings with defect $\rightarrow \kappa_d^F$. By (T_d^s) all $t_j^{(n)} < s_0$ eventually. In the proof of Theorem 5.1 the defect is at most $1 + P_d(\pi)$ if $x_0 > \pi$ (strictly below κ_d^F since $x_0^* < \pi$) and, by convexity in P , at most $\lambda + (1 - \lambda) \sup_\eta B_d(x_0, \eta)$ otherwise, where $\lambda = 1 - P/P_d(x_0) \in [0, 1]$. Since the defect tends to $\kappa_d^F > 1$, (W_d^s) and compactness give $\lambda^{(n)} \rightarrow 0$, $x_0^{(n)} \rightarrow x_0^*$ and maximizing directions $\eta \rightarrow 0$; $\lambda^{(n)} \rightarrow 0$ and strict concavity of $\log \varphi$ force the frequencies to the equal point t^* . Couplings with fixed marginals form a tight family; a limit coupling at t^* has defect κ_d^F , attained only in the direction $\eta = 0$: $P_d(x_0^*) - \mathbb{E} \cos S = \kappa_d^F$, i.e. $\mathbb{E} \cos S = \cos x_0^*$. Corollary 4.11 gives $\sum_j |X_j| \equiv dm$. Conversely, if $|X|$ is d -CM, Corollary 4.4 and Theorem 5.1(i) give $K_d^F = \kappa_d^F$. $\square$

Corollary 9.2 (Gaussian marginals). *For standard Gaussian marginals and every $d \geq 3$,*

$$K_d^G < \kappa_d^G = \max_x \{e^{-\pi x^2/(4d)} - \cos x\};$$

κ_d^G is an upper bound that is not attained.

Proof. $\varphi = e^{-t^2/2}$ has no zeros ($s^* = 0$) and $\log \varphi$ is strictly concave; $\gamma = 1/\pi$; (W_d^s) holds for all $d \geq 3$ (Appendix: brackets for $d \leq 19$; beyond, Lemma A.2 and $\kappa_{20}^G = 1.6926 > 1 - \cos x_c$ with Lemma A.1; strict exactness at x_0^* , e.g. $\gamma \sin x_0^*/(x_0^* P_3(x_0^*)) = 0.168 < 4/\pi^2$). Since $|X|$ is unbounded, $\sum |X_j|$ cannot be constant. $\square$

Proposition 9.3 (the mixability conjecture fails for $d = 3$). *Let $\mathbb{P}(X = 0) = \mathbb{P}(X = 1) = \mathbb{P}(X = -1) = \frac{1}{3}$. Then $|X|$ is 3-CM (one zero at a uniformly random place) and $\kappa_3^F = (5\sqrt{5} - 7)/4 = 1.0450849719$, but $K_3^F \geq 1.0673306671$.*

Proof. With probability $\frac{1}{3}$ let $X_1 = X_2 = X_3 = 0$; otherwise let $\varepsilon = \pm 1$ be a fair sign and $X_3 = \varepsilon$, $X_1 = X_2 = -\varepsilon$; each X_j has law F . At $t = (\frac{2\pi}{5}, \frac{2\pi}{5}, \frac{5\pi}{6})$, $S = \varepsilon(t_3 - t_1 - t_2) = \varepsilon \frac{\pi}{30}$ off the event $\{X_1 = X_2 = X_3 = 0\}$, so $\mathbb{E} e^{iS} = \frac{1}{3} + \frac{2}{3} \cos \frac{\pi}{30}$, while $\prod \varphi(t_j) = \varphi(\frac{2\pi}{5})^2 \varphi(\frac{5\pi}{6}) < 0$ with $\varphi(t) = \frac{1}{3} + \frac{2}{3} \cos t$; the difference is 1.0673306671. $\square$

Outside the unimodal class, attainment of the formula is therefore not characterized by mixability; a compensating coupling with frequencies of different sizes beats it. For $d \geq 4$ the question is open (Section 12).

10. A NEW PHENOMENON WITHOUT SYMMETRY

For a law without symmetry, the fusions of Section 4 at a split $p \neq \frac{1}{2}$ put an imaginary part into the fused point, and this can increase the defect. For equal frequencies $t > 0$ (by conjugation, $t < 0$ gives nothing new) let $D(t, p) = |p e^{i dt \mu_-(p)} + (1 - p) e^{i dt \mu_+(p)} - \varphi(t)^d|$, where $\mu_\mp(p)$ are the means of the lower and upper p -parts.

Theorem 10.1. *If F has a concave density on a bounded interval, then $K_d^F \geq \sup_{t,p} D(t, p)$ for every $d \geq 3$.*

Proof. Both parts have concave densities, hence are d -completely mixable for $d \geq 3$ (Puccetti–Wang–Wang, Thm 4.2); apply Theorem 4.3. $\square$

For Beta(1, 2) (density $2(1 - x)$ on $[0, 1]$) and $d = 3$: $\sup D = 1.0726392817$ at $t^* = 5.18958$, $p^* = 0.43813$, while the best symmetric split $p = \frac{1}{2}$ gives 1.0681. The optimal two-point split is asymmetric.

Conjecture 10.2. *If F has a concave density on a bounded interval (possibly with analogues of the conditions of Theorem 5.1), then $K_d^F = \sup_{t,p} D(t, p)$ for $d \geq 3$.*

Evidence: at t^* the comonotone relaxation lies within $4 \cdot 10^{-7}$ of $D(t^*)$ (explicit two-tangent certificates in 3600 directions) for Beta(1, 2) ($d = 3, 4$) and an asymmetric triangular law ($d = 3$). The upper-bound tools (an equal-variable reduction of frequency splits, a monotone-path principle) will appear in a sequel.

11. WITHOUT SYMMETRY: THE EXTREMAL UNIMODAL MARGINAL

By Theorem 7.1, for large d the least restrictive marginal minimizes $\text{Var}X/\tau^2$, $\tau = \mathbb{E}|X - \text{med}X|$; among symmetric unimodal laws the minimum is $\frac{4}{3}$, attained by the uniform law. Without symmetry it is smaller (Remark 8.5), and the uniform law is not extremal at any $d \geq 3$.

Unimodality. X is unimodal about M if its distribution function is convex on $(-\infty, M)$ and concave on (M, ∞) (an atom at M is allowed); M is then a mode. By Khinchine's theorem [8, 7] this holds iff $X = M + UZ$ with $U \sim U[0, 1]$ independent of Z .

Proposition 11.1 (the 9/7 inequality). *Let X be unimodal with $0 < \text{Var}X < \infty$ and $\tau = \mathbb{E}|X - \text{med}X|$.*

- (i) $\text{Var}X \geq \frac{9}{7} \tau^2$, with equality if and only if X is an affine image of $\frac{1}{4}\delta_0 + \frac{3}{4}U[0, 1]$.
- (ii) *If some median of X is a mode of X (in particular, if X is symmetric), then $\text{Var}X \geq \frac{4}{3} \tau^2$, with equality if and only if X is uniform.*

With $\text{ES}_{1/2}(X) = 2 \int_{1/2}^1 F^{-1}(u) du$ one has $\tau = \text{ES}_{1/2}(X) - \mathbb{E}X$, so (i) is the case $p = \frac{1}{2}$ of the worst-case expected shortfall over unimodal laws with given mean and variance, which is known in closed form: Li, Shao, Wang and Yang [9] treat Range Value-at-Risk under mean, variance and unimodality; see also [10, 11], and [12, Remark 8] for the formula $\sup(\text{ES}_p - \mathbb{E}X) = \sigma \sqrt{p(8-9p)/(3(1-p))}$, $p \leq \frac{1}{2}$, which equals $\frac{\sqrt{7}}{3}\sigma$ at $p = \frac{1}{2}$. We include a short proof by the two-point principle; it also gives the uniqueness of the extremal law and part (ii), which we have not found stated. Part (ii) does not follow from the Gauss–Winckler inequality $\mathbb{E}(X - M)^2 \geq \frac{4}{3}(\mathbb{E}|X - M|)^2$ about a mode M [13], since $\text{Var}X = \mathbb{E}(X - M)^2 - (\mathbb{E}X - M)^2$.

Lemma 11.2 (two-point decomposition). *Let ν be a probability law on $\mathbb{R}$ and g measurable with values in $[0, 1]$, $\int g d\nu = c$. Then ν is a mixture of laws with at most two atoms, each with $\int g = c$.*

Proof. Let μ be the law of $g(Z)$, $Z \sim \nu$, with restrictions $\mu_{\pm}$ to $\{y \gtrless c\}$ and $D = \int (y - c) d\mu_+ = \int (c - y') d\mu_-$. Then

$$\mu = \mu(\{c\})\delta_c + \iint \frac{(c - y')\delta_y + (y - c)\delta_{y'}}{y - y'} \cdot \frac{(y - y')\mu_+(dy)\mu_-(dy')}{D},$$

a mixture of two-point laws with mean c (the marginals are μ_+ and μ_-). Disintegrating ν along $g(Z)$ and drawing Z independently from the two conditional laws lifts each two-point law of $g(Z)$ to a mixture of two-point laws of Z with the same g -mean. $\square$

Proof of Proposition 11.1. Reduction. Fix a mode M of X and write $X = M + UZ$. The ratio $\text{Var}X/\tau^2$ is invariant under $X \mapsto \alpha X + \beta$ ($\alpha \neq 0$), so let $M = 0$. Then $\mathbb{E}X = \mathbb{E}Z/2$, $\mathbb{E}X^2 = \mathbb{E}Z^2/3$, and X has no atom except possibly at 0.

Median at the mode. Suppose 0 is a median of X ; this is the situation of (ii) (a symmetric unimodal law has its centre as a mode and a median). Then $p_{\pm} = \mathbb{P}(\pm Z > 0) = \mathbb{P}(\pm X > 0) \leq \frac{1}{2}$. With $a = \mathbb{E}Z_+$, $b = \mathbb{E}Z_-$: $\mathbb{E}Z_{\pm}^2 \geq (\mathbb{E}Z_{\pm})^2/p_{\pm} \geq 2(\mathbb{E}Z_{\pm})^2$ by Cauchy–Schwarz, so $\text{Var}X \geq \frac{2}{3}(a^2 + b^2) - \frac{1}{4}(a - b)^2$ and $\tau = \frac{a+b}{2} > 0$; with $a + b = 1$, $a = s$, the ratio is at least $4(\frac{1}{3}s^2 - \frac{1}{3}s + \frac{5}{12}) = \frac{4}{3} + \frac{4}{3}(s - \frac{1}{2})^2 \geq \frac{4}{3}$. Equality forces $s = \frac{1}{2}$ and equality in both Cauchy–Schwarz steps, i.e. $\mathbb{P}(Z > 0) = \mathbb{P}(Z < 0) = \frac{1}{2}$ with Z constant on each event; so $Z = \pm c$ with probability $\frac{1}{2}$ each and X is uniform. This proves (ii), and (i) when a mode is a median, since $\frac{4}{3} > \frac{9}{7}$.

Median off the mode. By reflection and scaling let the median be 1. Since X has no atom at 1, $\mathbb{P}(X \leq 1) = \frac{1}{2}$. For a value z of Z put $\psi(z) = \mathbb{P}(Uz \leq 1)$ ($= 1$ for $z \leq 1$, $= 1/z$ for $z \geq 1$) and $\phi(z) = \mathbb{E}|Uz - 1|$ ($= 1 - z/2$ for $z \leq 1$, $= z/2 - 1 + 1/z$ for $z \geq 1$). With ν the law of Z : $\int \psi d\nu = \frac{1}{2}$, $\tau = \int \phi d\nu$, and the claim is

$$H(\nu) := \frac{7}{3} \int z^2 d\nu - \frac{7}{4} \left(\int z d\nu \right)^2 - 9 \left(\int \phi d\nu \right)^2 \geq 0.$$

By Lemma 11.2 with $g = \psi$, $\nu = \int \nu_s \pi(ds)$ with two-point ν_s satisfying $\int \psi d\nu_s = \frac{1}{2}$; since $(\int L(\nu_s) d\pi)^2 \leq \int L(\nu_s)^2 d\pi$ for linear L , $H(\nu) \geq \int H(\nu_s) \pi(ds)$. So let ν have a left atom $a \leq 2$ (probability $1 - q$) and a right atom $b \geq 2$ (probability q); the one-point law $Z \equiv 2$ gives $H = \frac{1}{12}$.

Case $a \leq 1$. Here $q = \frac{b}{2(b-1)}$; H is quadratic in a with leading coefficient $\frac{(b-2)(b+5)}{6(b-1)^2} \geq 0$ and minimizer $a^*(b) = -\frac{3(b-3)(b+3)}{4(b+5)} \leq 1$ (equivalently $(3b+7)(b-1) \geq 0$), and

$$\min_a H = \frac{7(b-3)^2(b+6)}{96(b+5)} \geq 0,$$

with equality only at $b = 3$, $a = 0$, $q = \frac{3}{4}$.

Case $1 \leq a \leq 2$. Here $q = \frac{b(2-a)}{2(b-a)}$ and $H = N(a, b)/12$ with $N = -12a^2b^2 + 34a^2b - 20a^2 + 34ab^2 - 95ab + 54a - 20b^2 + 54b - 27$. Then $N(a, 2) \equiv 1$, N is convex in b (leading coefficient $2(2-a)(6a-5) \geq 0$), and $\partial_b N(a, 2) = (2-a)(14a-13) \geq 0$; hence $N \geq 1$.

Equality in (i) forces $H(\nu_s) = 0$ for π -a.e. s , so every component, hence ν , is the law of $Z \in \{0, 3\}$ with probabilities $(\frac{1}{4}, \frac{3}{4})$; then X has an atom $\frac{1}{4}$ at 0 and is uniform on $[0, 3]$ otherwise, the image of $\frac{1}{4}\delta_0 + \frac{3}{4}U[0, 1]$ under $x \mapsto 3x$. Conversely, for $\frac{1}{4}\delta_0 + \frac{3}{4}U[0, 1]$: median $\frac{1}{3}$, $\tau = \frac{1}{4} \cdot \frac{1}{3} + \frac{3}{4} \cdot \frac{1/9+4/9}{2} = \frac{7}{24}$, $\text{Var} = \frac{1}{4} - \frac{9}{64} = \frac{7}{64}$, ratio $\frac{9}{7}$. $\square$

Corollary 11.3 (asymptotically extremal marginal). *For every non-degenerate unimodal F with bounded support, $\lim_{d \rightarrow \infty} d(2 - K_d^F) = \frac{\pi^2 \text{Var} X}{2\tau^2} \geq \frac{9\pi^2}{14}$, with equality iff F is an affine image of $\frac{1}{4}\delta_0 + \frac{3}{4}U[0, 1]$; if a median is a mode (in particular for symmetric F), the bound is $\frac{2\pi^2}{3}$, attained by the uniform law.*

Proof. Cramér's condition is automatic: with $p_0 = \mathbb{P}(Z = 0) < 1$, $\varphi(t) = e^{itM}(p_0 + (1-p_0)\psi(t))$, where ψ is the characteristic function of a mixture of uniform laws, hence absolutely continuous; so $|\varphi(t)| \rightarrow p_0 < 1$, and $|\varphi(t)| < 1$ for $t \neq 0$ since a law with an absolutely continuous component is not lattice. Apply Theorem 7.1 and Proposition 11.1. $\square$

For finite d we use the fusions $D(t, p)$ of Section 10 and the mixability criterion (M1) of Section 2.

Lemma 11.4 (atoms at the left end). *Let $G = w_0\delta_0 + (1-w_0)H$ with H having a nonincreasing density on $[0, b]$. If the mean of G is at least b/d , then G is d -completely mixable.*

Proof. Replace $w_0\delta_0$ by $w_0U[0, \varepsilon]$: the result G_ε has a nonincreasing density on $[0, b]$ and a larger mean, so it is d -CM by (M1). Complete mixes of G_ε are tight (support in $[0, b]$); a subsequential limit in law has marginals G , and its coordinate sum is the limit of the constants $d\mathbb{E}G_\varepsilon$. $\square$

Theorem 11.5 (finite d). *Let $F = \frac{1}{4}\delta_0 + \frac{3}{4}U[0, 1]$ and U the uniform law. Then $K_d^F > K_d^U$ for every $d \geq 4$. For $d = 3$, $F' = \frac{41}{250}\delta_0 + \frac{209}{250}U[0, 1]$ has $K_3^{F'} \geq 1.0878685$, which exceeds $K_3^U = 1.0849633$, $K_3^{F^*} = 1.0874849$ (Theorem 8.7) and the supremum 1.0874848690 of κ_3^F in Proposition 8.8.*

Proof of Theorem 11.5. Attainable fusions. For $F_w = w\delta_0 + (1-w)U[0, 1]$ and $p > w$, the lower p -part is $\frac{w}{p}\delta_0 + \frac{p-w}{p}U[0, q]$, $q = \frac{p-w}{1-w}$, with mean $\frac{(1-w)q^2}{2p}$; the upper part $U[q, 1]$ is d -CM for every $d \geq 2$ by (M1). By Lemma 11.4 the lower part is d -CM if $\frac{(1-w)q^2}{2p} \geq \frac{q}{d}$. For $w = \frac{1}{4}$, $p = \frac{1}{2}$: $q = \frac{1}{3}$, mean $\frac{1}{12} \geq \frac{1}{3d}$ for $d \geq 4$. For $w = \frac{41}{250}$, $p = \frac{123}{250}$, $d = 3$: mean $= \frac{82}{627} = \frac{q}{3}$ exactly. By Theorem 4.3, $K_d^{F_w} \geq D(t, p)$ for all $t > 0$.

$d = 3$ and $4 \leq d \leq 200$. $D(t, p)$ is an explicit expression ($\varphi_w(t) = w + (1-w)\frac{e^{it}-1}{it}$, rational $\mu_\pm$) and is bounded below at explicit t by interval arithmetic; $K_d^U = \kappa_d$ (Corollary 6.1) is bounded above by a grid with a Lipschitz bound (Appendix A.6). All margins are positive (Table 2).

$d \geq 200$. Let $t_d = \frac{24\pi}{7d}$. Then $dt_d\mu_- = \frac{2\pi}{7}$ and $dt_d\mu_+ = \frac{2\pi}{7} + 2\pi$ ($\mu_- = \frac{1}{12}$, $\mu_+ = \frac{2}{3}$), so the fusion equals $e^{2\pi i/7}$; and with $\mu = \frac{3}{8}$, $\psi(t) = e^{-i\mu t}\varphi(t)$: $\varphi(t_d)^d = e^{9\pi i/7}\psi(t_d)^d = -e^{2\pi i/7}\psi(t_d)^d$. Hence $D(t_d, \frac{1}{2}) = |1 + \psi(t_d)^d| \geq 1 + \text{Re} \psi(t_d)^d$. By Lemma A.4,

$$D(t_d, \frac{1}{2}) \geq 1 + e^{-a/d} - \frac{14.6}{d^2}, \quad \kappa_d \leq 1 + e^{-b/d} + \frac{9.98}{d^2}, \quad a = \frac{9\pi^2}{14}, \quad b = \frac{2\pi^2}{3}.$$

Since $1 - e^{-y} \geq ye^{-y}$, $d^2(e^{-a/d} - e^{-b/d}) \geq d\frac{\pi^2}{42}e^{-b/d} \geq 45.4 > 24.6$ for $d \geq 200$. $\square$

d	lower bound for K_d^F	upper bound for K_d^U	margin
3 (F')	1.0878685	1.0849652	0.0029 (vs. F^* : 0.00038)
4	1.2002921	1.1907241	0.0096
5	1.2914829	1.2796741	0.0118
8	1.4731083	1.4608883	0.0122
20	1.7381118	1.7301725	0.0079
60	1.9014265	1.8980328	0.0034
100			0.0022
200			0.0011

TABLE 2. Certified margins (interval arithmetic); $F = \frac{1}{4}\delta_0 + \frac{3}{4}U[0, 1]$, median split. The margin behaves like $\frac{\pi^2}{42d}$.

12. OPEN PROBLEMS

- (1) The supremum of K_3^F over symmetric unimodal laws with 3-CM $|X|$: conjecturally equal to $\sup \kappa_3^F = 1.0874848690$ (Proposition 8.8), which F^* approaches within $1.4 \cdot 10^{-9}$. Peaked limits of this class (an atom plus a spread-out part, $r = \varphi(t) \leq \frac{1}{3}$) give at most $1 + r^3 \leq 28/27$ by Theorem 3.1.
- (2) The three-point family $\mathbb{P}(X = 0) = w$, $\mathbb{P}(X = \pm 1) = \frac{1-w}{2}$ for $d \geq 4$: is $K_d = \kappa_d$ exactly when $|X|$ is d -CM?
- (3) The size of the Gaussian gap: numerically $K_3^G \approx 1.081 < \kappa_3^G = 1.0848$.
- (4) Conjecture 10.2; exact identities $K_d^F = \sup D$ for $d \geq d_F$.
- (5) Theorem 7.1 for non-lattice laws without Cramér's condition.
- (6) The exact value of K_d^F for $F = \frac{1}{4}\delta_0 + \frac{3}{4}U[0, 1]$ (is it $\sup D$?), and the extremal unimodal marginal at a fixed d : is it an atom plus a uniform part for every $d \geq 3$?

APPENDIX A. FINITE VERIFICATIONS

A.1. Method. Each remaining finite claim has the form $F \leq \kappa$ on a box, where F is built from elementary functions that are monotone in each variable on the box (or have explicit Lipschitz bounds). On a sub-box, an upper bound for F is obtained by evaluating every piece at the corner that maximizes it. Sub-boxes are bisected until each is accepted *by a margin* ($F < \kappa - 10^{-3}$) or *by exactness*: in the window, the inequality $\gamma \sin x_0 \eta^2 / x_0 \leq P(1 - \cos \eta)$ on the box, which forces the second term of B_d to be at most $P_d(x_0) - \cos x_0 \leq \kappa$; it is checked through $\gamma \sin x_a / (x_a P_d(x_b)) \leq (1 - \cos \eta_b) / \eta_b^2$. Lower bounds for κ are the values of the formula at explicit points. No interval library is used: margins are $\geq 10^{-3}$ against rounding errors of order 10^{-15} . In every box $\eta \geq \frac{\pi}{2}$ is trivial, since then $B_d \leq 1 + P \cos \eta \leq 1 < \kappa$.

A.2. Two general lemmas.

Lemma A.1 (monotonicity in d). *If $\log \varphi$ is concave on the first lobe, then $P_d(x) = \varphi(x/(dm))^d$ is nondecreasing in $d \geq 3$ for $0 \leq x \leq \pi$ ($\pi/(dm)$ lies in the lobe by Lemma 2.2). Consequently κ_d^F is nondecreasing, and exactness on the window, condition (T_d) and every inequality of the form $\kappa_d^F \geq c$ persist for all larger d once they hold.*

Proof. With $\psi = \log \varphi$ concave and $\psi(0) = 0$, $\psi(u)/u$ is nonincreasing, so $d\psi(x/(dm)) = \frac{x}{m} \cdot \frac{\psi(x/(dm))}{x/(dm)}$ is nondecreasing in d . Exactness is monotone in P . $\square$

Lemma A.2 (the region $x_0 < x_c$ for large d). *If $\gamma \leq \gamma_0$, then $B_d(x_0, \eta) \leq 1 - \cos x_c = x_c \sin x_c = 1.68915 \dots$ for all $0 < x_0 \leq x_c$ and $\eta \in [0, \pi]$, whatever $P_d(x_0) \in [0, 1]$. Hence this part of (W_d) holds as soon as $\kappa_d^F \geq 1 - \cos x_c$.*

Proof. For $x_0 \leq x_c$ the second term is $-1 + \sin x_c(x_0 + e) + P \cos \eta$ with $e = \min\{\gamma \eta^2 / x_0, \eta\}$, and $x_0 + e \leq \max\{x_c + \gamma \eta^2 / x_c, (1 + \gamma)\eta\}$. If the first entry is the larger, the second term is at most $x_c \sin x_c + \gamma \sin x_c \eta^2 / x_c - (1 - \cos \eta) \leq x_c \sin x_c$, since $(1 - \cos \eta) / \eta^2 \geq 2/\pi^2$ and $\gamma \leq \gamma_0$

(for $\cos \eta < 0$ the first term is already ≤ 1). Otherwise $\eta \geq x_c/(1 + \gamma) \geq 1.41$ and the first term is at most $1 + \cos 1.41 < 1.16$. $\square$

A.3. The verifications, law by law.

law	log φ concave	tail	window $[x_c, \pi]$	$x_0 < x_c$
uniform	analytic (sinc)	(T _d) for $d \geq 5$; $d = 4$: Lemma 5.5 ($\beta_{\max} = 0.762 < \beta_1 = 1.243$); $d = 3$: Lemma A.3	exactness at $d = 3$ by (C1), then all d by Lemma A.1	boxes $3 \leq d \leq 16$ (5–25 each, margin ≥ 0.005); $d \geq 17$ by Lemma A.2
triangular	analytic (sinc^2)	(T _d): $1 + 0.0472 < 1.0766$	boxes $3 \leq d \leq 19$; exactness at $d = 18$; $d \geq 19$ by Lemma A.1	boxes $3 \leq d \leq 20$ (ratio 0.273), then Lemma A.2
cubic B-spline	analytic (sinc^3)	(T _d): $s^* \leq 0.0103$	boxes $d = 3, 4, 5$; exactness for $d \geq 6$	boxes $3 \leq d \leq 19$; $d \geq 20$ by Lemma A.2
Epanechnikov	brackets ($\varphi\varphi'' - \varphi'^2 \leq -0.021$)	(T _d) for $d \geq 4$: $s^* \leq 0.0862$	boxes $d = 4$; exactness for $d \geq 5$	boxes $4 \leq d \leq 17$; $d \geq 18$ by Lemma A.2
F^* ($d = 3$)	brackets (≤ -0.949 on $[0, 0.599]$)	Lemma 5.5: $\beta_{\max} = 1.0891 < \beta_1 = 1.0932$; $s^{*2} < \tau$	25 boxes (18 by exactness)	29 boxes
Gaussian	analytic	none ($s^* = 0$)	boxes $d = 3, 4, 5$; exactness for $d \geq 6$ (ratio 0.889 at $d = 6$); strict at x_0^* (0.168, 0.126, 0.101 $< 4/\pi^2$)	boxes $3 \leq d \leq 19$; $d \geq 20$ by Lemma A.2

“Exactness for $d \geq d_1$ ” uses the crude form $\gamma \frac{\sin x_c}{x_c} \leq \frac{4}{\pi^2} P_d(\pi)$, i.e. exactness on the whole window for $\eta \leq \frac{\pi}{2}$ at once, then Lemma A.1. Mixability for the lower bound: uniform and triangular $d \geq 3$ ($b/m = 2, 3$), B-spline $d \geq 4$ ($b/m = 48/13$), Epanechnikov $d \geq 3$ ($b/m = 8/3$), F^* ($b/m = 3$).

A.4. The uniform tail for $d = 3$.

Lemma A.3. *For uniform marginals on $[-1, 1]$ and $d = 3$, if exactly one frequency $t_1 \in [\pi, 3\pi]$ lies beyond the lobe and $\beta = \frac{1}{2}(t_2 + t_3) \in [0.993, 1.599]$, the defect is at most κ_3 .*

Proof sketch. In every direction the defect is at most $1 - \frac{2}{\pi^2}(\rho(\psi) - \beta)_+^2 + Q \cos u$, where $\rho(\psi) = \mathbb{E} \text{dist}(t_1 X_1 - \psi, 2\pi\mathbb{Z})$ (Lemma 2.3 and Jensen), u is the distance of ψ from the dangerous direction and $Q \geq |P|$. Only $u \leq \arccos(\tau/Q)$ matters; $\partial_\psi \rho = 0$ at the dangerous direction by symmetry and $|\partial_\psi^2 \rho| \leq 1/t_1 \leq 1/\pi$, so $\rho \geq \rho^* - u^2/(2\pi)$ with ρ^* in closed form; the resulting lower bound is linear in $v = u^2$, so two closed-form inequalities at $v = 0$ and $v = u_0^2$ suffice. On 40 cells in (t_1, β) , with bounds from the monotonicity of ρ^* (quasi-concave), $|\text{sinc}|$ (unimodal on each lobe) and $\text{sinc}(\cdot)^2$, both inequalities hold with slack ≥ 0.0014 (table in the supplementary material). Below $\beta = 0.993$ Lemma 5.5 applies; above 1.599, $|P| \leq \tau$. For $t_1 \geq 3\pi$, $|\varphi(t_1)| \leq 1/(3\pi)$, so $|P| > \tau$ forces $\beta < 0.81$, while Lemma 5.5 settles $\beta \leq 1.12$. $\square$

A.5. Reproducibility. All certificates run in about 8 seconds (`run_all_certificates.py`, all passing); the 40-cell table and the scripts for F^* and the Gaussian law are in the supplementary material.

A.6. Certification for Section 11. Unlike the rest of this appendix, the comparison of Theorem 11.5 uses interval arithmetic. *Lower bounds.* $D(t, p)$ is evaluated with real interval arithmetic (mpmath, 40 digits): $\varphi_w(t) = w + (1 - w)\frac{\sin t}{t} + i(1 - w)\frac{1 - \cos t}{t}$, the power φ^d by repeated complex multiplication of intervals, and $|\cdot|$ by the square root of a sum of squares; the lower endpoint is reported.

Upper bounds for $\kappa_d = \max_x \{\text{sinc}(2x/d)^d - \cos x\}$. On $[0, \pi]$: a grid of step $h = 2 \cdot 10^{-6}$ ($d \leq 60$) or 10^{-6} ($61 \leq d \leq 200$) and $|g'| \leq 2 \max |\text{sinc}'| + 1 < 1.88$. For $x > \pi$: $|\text{sinc}(2x/d)| \leq \text{sinc}(2\pi/d)$

while $2x/d \leq \pi$, and $|\operatorname{sinc}| \leq 0.2173$ beyond, so the value is at most $1 + \operatorname{sinc}(2\pi/d)^d$, the value at π .

Lemma A.4 (tail constants, $d \geq 200$). *With $t_d = \frac{24\pi}{7d}$: $D(t_d, \frac{1}{2}) \geq 1 + e^{-a/d} - 14.6/d^2$ and $\kappa_d \leq 1 + e^{-b/d} + 9.98/d^2$.*

Proof. Lower. $Y = X - \frac{3}{8} \in [-\frac{3}{8}, \frac{5}{8}]$, $\mathbb{E}Y^2 = \sigma^2 = \frac{7}{64}$, $\mathbb{E}|Y|^3 \leq \frac{5}{8}\sigma^2$. From $|e^{iy} - 1 - iy + \frac{y^2}{2}| \leq \frac{|y|^3}{6}$: $\psi(t) = 1 - \frac{\sigma^2 t^2}{2} + R$, $|R| \leq r := \frac{35}{3072}|t|^3$. With $u = \psi - 1$, $|u| \leq \frac{\sigma^2 t^2}{2} + r \leq \frac{1}{2}$ and $|\log(1+u) - u| \leq |u|^2$, so $d \log \psi(t_d) = -\frac{a}{d} + de_1$ with $d|e_1| \leq d(r + |u|^2) \leq \frac{14.24}{d^2} + \frac{41.2}{d^3} \leq \frac{14.45}{d^2}$ ($\frac{\sigma^2 t_d^2}{2} = \frac{a}{d^2}$, $dr = 14.24/d^2$). Then $\operatorname{Re} \psi^d \geq e^{-a/d-\epsilon} \cos \epsilon \geq e^{-a/d}(1 - \epsilon - \epsilon^2/2)$ with $\epsilon = d|e_1|$.

Upper. On $[0, \pi]$, $\operatorname{sinc} y \leq e^{-y^2/6}$ ($|y| < \pi$; $\log \operatorname{sinc} y = \sum_k \log(1 - \frac{y^2}{k^2 \pi^2}) \leq -\frac{y^2}{6}$) gives $g(x) \leq e^{-\alpha x^2} - \cos x$, $\alpha = \frac{2}{3d}$. For $x = \pi - \delta$, $\delta \leq 1$: $e^{-\alpha(\pi-\delta)^2} \leq e^{-\alpha\pi^2} + 2\alpha\pi e^{2\alpha\pi}\delta$ and $-\cos x = \cos \delta \leq 1 - \frac{11}{24}\delta^2$, so $g \leq 1 + e^{-b/d} + \frac{6}{11}(2\alpha\pi e^{2\alpha\pi})^2 \leq 1 + e^{-b/d} + 9.98/d^2$. For $\delta \geq 1$, $g \leq 1 + \cos 1 < 1.55 < 1 + e^{-b/d}$; for $x > \pi$ see above. $\square$

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